LAB 19.3

Name : Devendar

Roll.No : 2303A53036
Batch-46

Task 1: Python → C++

Prompt

Convert a Python Student class into equivalent C++ code with constructor and methods.

끲 Python Code

```
class Student:     def
__init__(self, name, age):
        self.name = name
        self.age = age

    def display(self):
        print(self.name, self.age)

s = Student("varun", 20)
s.display()
```

끲 Output

varun 20

끲 C++ Code

```
#include <iostream>
using namespace std;

class Student {
string name;
    int age;

public:
    Student(string n, int a) {
name = n;
        age = a;
```

```
    }

    void display() {
        cout << name << " " << age;
    }
};

int main() {     Student
s("varun", 20);
s.display(); }
```

Output varun

20

Task 2: Java → Python (Prime)

Prompt

Convert a Java isPrime() method into a Python func on.

Code

```
def isPrime(n):     if
n <= 1:         return
False     for i in
range(2, n):         if n
% i == 0:
return False
    return True

print(isPrime(7))
print(isPrime(10))
```

Output

True
False

Task 3: Pseudocode → Python

Prompt

Convert sor ng pseudocode into Python (bubble sort).

Code

```
arr = [5, 2, 9, 1]

for i in range(len(arr)):
    for j in range(len(arr)-i-1):
if arr[j] > arr[j+1]:
            arr[j], arr[j+1] = arr[j+1], arr[j]

print(arr)
```

Output

```
[1, 2, 5, 9]
```

Task 4: SQL → Pandas

Prompt

Convert SQL query into Pandas: select name and salary where salary > 50000.

Code

```
import pandas as pd

data = {
    'name': ['A', 'B', 'C'],
    'salary': [40000, 60000, 70000]
}

df = pd.DataFrame(data)

result = df[df['salary'] > 50000][['name', 'salary']]
print(result)
```

Output

	name	salary
1	B	60000
2	C	70000

Task 5:
Python → C
(Search)

Prompt

Convert a Python linear search algorithm into C.

Python Code

```
arr = [1, 3, 5, 7]
key = 5

for i in range(len(arr)):     if
arr[i] == key:
print("Found at index", i)
```

Output

Found at index 2

C Code

```
#include <stdio.h>

int main() {     int
arr[] = {1,3,5,7};
    int key = 5;

    for(int i=0;i<4;i++){
if(arr[i]==key){              prin
("Found at index %d", i);
        }
    }
}
```

Output

Found at index 2